

CHAPTER – 11

GENERIC IN COLLECTIONS FRAMEWORK

GENERIC IN JAVA: -

Generics allow Java classes, methods, and collections to work with type safety by specifying the data type at compile time.

SYNTAX: -

```
ArrayList<String> list = new ArrayList<>();
```

TYPES: -

Generic class

Generic methods

CUSTOM OBJECT WITH GENERICS:-

A custom object is an object created from a user-defined class.

A custom object is an object created from a class that YOU define, not a built-in class like String or Integer.

EXAMPLE: -

```
class Student<T> {
```

```
T data;
```

```
Student(T data) {  
  
    this.data = data;  
  
}  
  
void display() {  
  
    System.out.println(data);  
  
}  
  
}  
  
public class GenericExample {  
  
    public static void main(String[] args) {  
  
        Student<String> s1 = new Student<>("Shalini");  
  
        Student<Integer> s2 = new Student<>(101);  
  
        s1.display();  
  
        s2.display();  
  
    }  
  
}
```

SORTING IN COLLECTIONS: -

Sorting in collections means arranging elements in a specific order (ascending or descending).

In Java, collections can be sorted using: Comparable (natural sorting)

Comparator (custom sorting logic)

COMPARABLE:

- Comparable is an interface used to define the natural sorting order of objects.
- A class implements Comparable and overrides the compareTo() method.

EXAMPLE:-

```
import java.util.*;

class Student implements Comparable<Student> {

    int id;

    Student(int id) {

        this.id = id;

    }

    public int compareTo(Student s) {

        return this.id - s.id;

    }

}

public class ComparableDemo {

    public static void main(String[] args) {

        ArrayList<Student> list = new ArrayList<>();

        list.add(new Student(3));

        list.add(new Student(1));
```

```
list.add(new Student(2));
```

```
Collections.sort(list);
```

```
for (Student s : list) {
```

```
    System.out.println(s.id);
```

```
}
```

```
}
```

```
}
```

COMPARATOR: -

- Comparator is an interface used to define custom or multiple sorting orders.
- Sorting logic is written outside the class.